IRoad Control Panel PCS - Part 1

Document ID: CP-PCS-P1 **Title:** Super Admin Control Panel - Master Policy, Architecture, and Scope **Phase:** 1 (MVP)

1. Executive Summary & Design Philosophy

The **Super Admin Control Panel (CP)** is the central command center for the IRoad SaaS platform. It is strictly isolated from the Tenant (Transporter) workspace, serving as the core engine for operational governance, financial reconciliation, and platform security.

Core Philosophies for Phase 1:

1. **Isolated Execution vs. Centralized Governance:** While every Tenant operates within an isolated database schema for their daily transport operations, the Super Admin Panel governs from a Central Master Database. Shared services (like Support Tickets and Billing) are managed centrally and exposed to Tenants via secure API Bridges.
2. **Authority-Driven State Machine:** For shared entities (e.g., Subscription Renewals, Support Tickets), the Tenant can initiate requests or queries, but the Super Admin holds the absolute authority to transition critical states (e.g., Approving, Closing, Suspending).
3. **Zero-Touch Automation & Resilience:** The architecture relies heavily on background automation (Pro-Rata calculations, automated recurring billing, geo-based tax routing) while providing manual fallback mechanisms (Offline Bank Transfers) to ensure business continuity.
4. **Absolute Data Integrity:** Global data (Countries, Currencies, Tax Codes) is strictly managed by the CP to prevent data pollution across isolated tenant databases.

2. Scope Boundaries (Phase 1)

To ensure a successful, secure, and scalable MVP launch, strict boundaries are established regarding what is included and excluded in the CP capabilities.

2.1 In-Scope (Included)

- **Static Roles & Dynamic Accounts (P3):** Fixed system roles (Super Admin, Sales, Support) assigned to staff accounts with strict suspension protocols.
- **Tenant CRM & Provisioning (P5):** Centralized directory for Tenant profiles, automated onboarding, and the enforcement of the "Kill Switch" for instant suspension.
- **Subscription & Product Engine (P4):** Dynamic configuration of Base Plans, Add-ons, resource hard-limits (e.g., Max Trucks, Users), and Promo-Code generation.
- **Dynamic Payment Infrastructure (P7):** Centralized vault for Payment Gateway API keys and dynamic routing of Payment Methods (Online/Offline) based on selected currencies.
- **Pro-Rata Auto-Billing (P5):** Automated calculation engine for upgrades and renewals mid-cycle to ensure zero revenue leakage.
- **Advanced Tax & FX Engine (P9):** Geolocation-based tax routing (applying standard vs. zero-rated taxes based on Tenant IP/Profile) and an immutable FX rate snapshot system for standardizing platform revenue reports.
- **Communication & Alerts Engine (P6):** Centralized SMTP/SMS configuration, dynamic notification templates, fallback routing, and Push Notification broadcasting.
- **Centralized Support Help Desk (P10):** A unified ticketing inbox residing in the Master DB, equipped with internal notes and canned responses, communicating securely with Tenants via an API bridge.
- **Security & Active Session Control (P11):** Independent TTL policies for Admins vs. Tenants, active JWT session monitoring, and real-time token revocation (Mass Revoke).
- **Immutable Audit Logging (P11):** A "Black Box" master system log capturing every administrative change with Old/New JSON payloads.

2.2 Out-of-Scope (Excluded / Deferred to Phase 2)

- 🔴 **Impersonation / Login-As Tenant:** The ability for Super Admins to log into a Tenant's workspace using a backdoor token is **strictly excluded** in Phase 1 to minimize privacy liability and simplify the security architecture.
- 🔴 **Affiliate / Referral System:** Partner dashboards and referral commission tracking are deferred.

- 🔴 **Deep White-Labeling:** Tenants cannot change system colors, domains, or UI themes (only logo and basic profile data are allowed).

3. Architectural Orientation: Module Typology

The CP is logically divided into two distinct module types based on their **Interaction Scope** and Database placement.

Type A: Exclusive CP Modules (Isolated Scope)

These modules belong entirely to the IRoad administration and reside exclusively in the Master DB. Tenants have zero visibility or interaction with these components.

- *Orientation:* One-way configuration and internal management.
- *Examples:*
  - **Security & Session Policies (P11):** TTL rules, brute-force limits, and access logs.
  - **Payment & Communication Configs (P6, P7):** API keys, Webhook secrets, and Event Mapping.
  - **Global Master Data & Tax (P8, P9):** Base currencies, exchange rates, and dynamic tax codes.
  - **Subscription Plans Master (P4):** The underlying blueprint for limits and pricing.

Type B: Interactive & Shared Modules (Bridged Scope)

These modules act as the bridge between the Tenant's isolated environment and the Super Admin's Central DB.

- *Orientation:* Request-and-Approve, API Bridging, and Snapshotting.
- *Examples:*
  - **Support Ticketing System (P10):** The actual tickets live in the Master DB. The Tenant views/replies to them via API requests carrying their `Tenant_ID`. The Admin retains absolute authority to assign, reprioritize, or force-close tickets.
  - **Subscription Orders & Billing (P5):** The Tenant views their active plan and initiates an Upgrade/Renewal. If via Gateway, the system auto-approves. If via Bank Transfer (Managed Billing), the Order enters a `Pending_Payment` state requiring Admin verification of the attached receipt before generating the Immutable Standard Invoice.

4. Setup & Provisioning Mechanism

How a new Tenant (Company) is introduced to the IRoad ecosystem:

1. **The Commercial Event:** IRoad Sales signs a contract, or the client subscribes via the public portal.
2. **System Provisioning:** The Super Admin (or automated webhook) creates a new `Tenant Master Record` (P5).
3. **Plan Assignment & Generation:** The system attaches a `Subscription Plan` and automatically:
   - Generates a unique `Tenant ID`.
   - Provisions an isolated database schema for the Tenant.
   - Seeds the Tenant's DB with required Global Master Data (Countries, Currencies).
   - Creates the default Super Admin user account for the Tenant.
4. **Handover:** The Communication Engine (P6) dispatches a secure welcome email with login credentials to the Tenant's administrator.

5. Security & Audit Policy

Given the absolute authority of the Super Admin Panel, security is governed by three unyielding pillars (detailed in P11):

- **1. The "Kill Switch" Execution:** Changing a Tenant's status to `Suspended` (due to violation or non-payment) triggers an immediate system-wide token revocation. All active sessions for that Tenant's web users and mobile drivers are destroyed instantly via Redis, enforcing an absolute lockout.
- **2. The Black Box (Master Audit Log):** Every Create, Update, Delete, or Status Change performed by an IRoad Admin on any configuration or commercial record is logged immutably. The database user executing the backend application has NO `UPDATE` or `DELETE` privileges on the Audit Log table itself.
- **3. Transactional Immutability (Snapshots):** Once a financial transaction (Standard Invoice or Order) is executed, all related lookup data (Tax Rates, FX Rates, Prices, Client Names) is snapshotted (hard-copied) into the transaction record. Subsequent changes to Master Data will never alter historical financial records.

(End of CP-PCS-P1)

Control Panel PCS Part 2

Document ID: CP-PCS-P2 **Title:** Super Admin Control Panel - Master Menu & Navigation Tree **Phase:** 1 (MVP)

> **Note:** This tree adopts the comprehensive and finalized management structure for the Super Admin, accurately reflecting all the developed configurations, master files, and workflows from `Part 3` to `Part 11`.

1. Dashboard (لوحة القيادة)

- **Overview:** Executive KPIs (MRR, Active Tenants, Growth).
- **Attention Center:** Pending actions, overdue support tickets, failed payments.
- **Quick Metrics:** System-wide usage stats (Total Trucks, Total Drivers).

2. System Users Management `[P3]`

- **Users Analytics Dashboard:** Behavioral tracking and staff productivity.
- **Admin Users Master:** Management of IRoad internal staff accounts.
- **Roles Master:** System roles definition (Super Admin, Sales, Support).

3. Subscription Plans Management `[P4]`

- **Subscription Plans Master:** Plan definitions, module toggles, and base cycle pricing.
- **Global Add-ons Pricing:** Centralized pricing for extra resources (users, trucks, storage).
- **Promo-Codes Master:** Marketing discount codes and limits.

4. Subscribers CRM & Billing `[P5]`

- **Subscribers List (Tenant Master):** Complete registry of all tenants and their current status.
- **Internal CRM Notes:** Sales and support activities timeline per tenant.
- **Subscription Orders:** Sales engine for New, Renewal, Upgrade, and Add-on orders.
- **Transactions Ledger:** Centralized wallet and manual/automated fund movements.
- **Standard Invoices:** Immutable, API-ready generated invoices.

5. Communication & Alerts `[P6]`

- **Gateways Settings:** SMTP & SMS API credentials.
- **Notification Templates:** Multilingual Email/SMS content blocks.
- **Events Mapping Engine:** Intelligent routing of system triggers to templates (with fallback).
- **Push Notifications:** Manual broadcasts and automated app alerts.
- **System Banners:** Global in-app announcements for tenants.
- **Internal Alerts Routing:** Dispatching critical system events to specific admin roles.

- **Communication Logs:** Immutable delivery audit trail.

6. Payment Infrastructure `[P7]`

- **Payment Gateways API:** Secure dynamic credentials for online providers (e.g., Stripe/Moyasar).
- **IRoad Bank Accounts:** Active bank accounts for offline transfers.
- **Payment Methods Configuration:** Dynamic checkout routing engine based on currency and type.

7. Global Master Data `[P8]`

- **Countries Master:** ISO codes, calling codes, and active statuses.
- **Currencies Master:** ISO currency codes, symbols, and precision.

8. System Configurations & Tax `[P9]`

- **IRoad Legal Identity:** Corporate details and logo for standard invoicing.
- **Global System Rules:** Timezones, billing cycles, and grace periods.
- **Tax Codes Master:** Dynamic geolocation-based tax routing.
- **Billing & Tax Settings:** Tax inclusivity and location verification logic (Audit/Enforce).
- **IRoad Base Currency:** Immutable accounting root currency.
- **Exchange Rates Manager:** FX rates against the Base Currency.
- **FX Rate Change Log:** Audit trail for currency value changes.

9. Support Management `[P10]`

- **Support Categories:** Dynamic ticket classifications.
- **Canned Responses:** Standardized quick-reply templates for agents.
- **Support Inbox (Tickets Master):** Centralized workspace for tenant issues.
- **Ticket Thread & Internal Notes:** Conversations including hidden admin-to-admin notes.

10. Security & Audit `[P11]`

- **Admin Security Settings:** TTL and lockout rules for IRoad staff.
- **Tenant Security Settings:** TTL and lockout rules for web users and mobile drivers.
- **Active Sessions Monitor:** Live JWT tracking with "Kill Switch" capabilities.
- **Authentication Log:** Brute-force detection and login history across all domains.
- **Master System Audit Log:** The immutable "Black Box" tracking all admin data modifications (Old vs New JSON).

(End of CP-PCS-P2)

Control Panel PCS Part 3

Document ID: CP-PCS-P3 **Title:** System Users Management & Analytics **Phase:** 1 (MVP) **Module:** 1. System Users Management

1.1 System Users Analytics Dashboard

1.1.1 Dashboard Identification Card

Attribute	Description
Dashboard ID	`DSH-CP-01-01`
Dashboard Name	System Users Analytics Dashboard
Dashboard Type	Analytical / Reporting (Read-Only)
Related Entities	`Admin Users`, `Roles`, `Tenants`, `Support Tickets`
Purpose	A specialized analytical dashboard strictly focused on tracking the behavior, activity levels, and distribution of the internal IRoad staff (Admin Users) to monitor team productivity and operational hygiene.

1.1.2 Dashboard Components & Metrics

A. Top-Level KPIs (مؤشرات الأداء السريعة):

- **Total Registered Staff:** Total count of all admin accounts (Excluding 'Suspended').
- **Currently Active Sessions:** Live counter of admins currently online (Active JWT tokens).
- **Suspended Accounts:** Total count of suspended staff accounts.
- **2FA Adoption Rate:** Percentage of active admins with `two_factor_enabled = True`.

B. Behavioral Analytics (تحليلات السلوك والدخول):

- **Login Activity Heatmap:** Visual chart showing login density across weekdays and hours.
- **Stale Accounts:** List/Count of admins with `last_login_at` older than 30 days.
- **Average Session Duration:** Calculated average time spent logged into the system per session.

C. Cross-Module Performance Metrics (مؤشرات الأداء التشغيلي حسب الدور): *(Note: Data aggregated based on `admin_id` linked to other modules).*

- **Sales Activity:** Number of new Tenants registered/activated by each Sales Rep current month.
- **Support Efficiency:** Number of tickets closed/replied to by each Support Agent.
- **Critical Actions Executed:** Count of sensitive actions (e.g., Plan downgrades, refunds) logged in the Audit Trail per admin.

D. Role Distribution (توزيع الأدوار):

- **Donut Chart:** Visual distribution of the team by `role_id` (e.g., 10% Super Admins, 40% Sales).

1.2 Admin Users Master (Platform Staff Accounts)

1.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-01-01`
Form Name	Admin Users Master
Form Type	Master File (Read/Write)
Related Entities	`Roles Master`
Purpose	Manages the internal staff accounts of the IRoad company. Strictly isolated from Tenant users. Forms the basis of Platform RBAC (Role-Based Access Control).

1.2.2 Main Fields (Data Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`admin_id`	UUID (PK)	Yes	Auto	Hidden. Primary Key.
`first_name`	VARCHAR(50)	Yes	None	Text Input.
`last_name`	VARCHAR(50)	Yes	None	Text Input.
`email`	VARCHAR(100)	Yes	None	Email Input. **Strictly UNIQUE** across the entire Control Panel DB.
`phone_number`	VARCHAR(20)	No	Null	Phone Input. International format.
`role_id`	UUID (FK)	Yes	None	Dropdown. Fetched from active `Roles Master`.
`status`	ENUM	Yes	'Pending'	Dropdown. Values: `Pending_Activation`, `Active`, `Suspended`.
`two_factor_enabled`	BOOLEAN	Yes	False	Toggle/Read-only flag. Shows if user completed 2FA setup.
`last_login_at`	TIMESTAMP	No	Null	Read-only. Auto-updated upon successful login.
`created_by`	UUID (FK)	Yes	Auto	Hidden. The Admin ID who created this account.
`updated_by`	UUID (FK)	No	Null	Hidden. The Admin ID who last modified this account.
`created_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.
`updated_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.

(Note: Passwords are NOT stored in plain text. A secure hash mechanism is managed purely by the backend auth controller).

1.2.3 Business Rules & Logic

- **A. The Invite Protocol:** The Super Admin does **not** set a password. Upon saving, status is `Pending_Activation`. System sends a secure 24-hour expiry invite link. The new staff sets their own password, and status changes to `Active`.
- **B. The "Kill Switch":** Changing status to `Suspended` immediately revokes all active JWT tokens for that user. Subsequent navigation forcefully redirects to login with an error.
- **C. Root Protection:** The primary system owner account (Admin ID #1) cannot have its `status` changed to `Suspended` or its `role_id` demoted. Hardcoded failsafe.

1.3 Roles Master (System Roles)

1.3.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-01-02`
Form Name	Roles Master
Form Type	Master File (Read/Write)
Related Entities	`Admin Users Master`, `Permissions Matrix` (Future Phase)
Purpose	Defines high-level operational groupings for IRoad staff. Acts as the container for permissions.

1.3.2 Main Fields (Data Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`role_id`	UUID (PK)	Yes	Auto	Hidden. Primary Key.
`role_name_en`	VARCHAR(50)	Yes	None	Text Input (English). **UNIQUE**.
`role_name_ar`	VARCHAR(50)	Yes	None	Text Input (Arabic). **UNIQUE**.
`description`	VARCHAR(255)	No	Null	Text Area. Brief explanation of the role.
`is_system_default`	BOOLEAN	Yes	False	Read-only flag. Indicates core system roles (e.g., Root Admin).
`status`	ENUM	Yes	'Active'	Dropdown. Values: `Active`, `Inactive`.
`created_by`	UUID (FK)	Yes	Auto	Hidden. The Admin ID who created this role.
`updated_by`	UUID (FK)	No	Null	Hidden. The Admin ID who last modified this role.
`created_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.
`updated_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.

1.3.3 Business Rules & Logic

- **A. Immutability:** Any role flagged as `is_system_default = True` cannot be deleted, modified in core properties, or set to `Inactive`.
- **B. Relational Safety (Deletion Constraint):** A Role cannot be deleted if there is at least one active Admin User currently assigned to it. It must be set to `Inactive` instead (hiding it from future assignment dropdowns without breaking historical records).

(End of CP-PCS-P3)

IRoad Control Panel PCS - Part 4

Document ID: CP-PCS-P4 **Title:** Subscription Plans Management **Phase:** 1 (MVP) **Module:** 2. Subscription Plans Management

2.1 Subscription Plans Master (Product, Limits & Pricing)

2.1.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-02-01`
Form Name	Subscription Plans Master
Form Type	Master File (Header-Detail)
Related Entities	`Subscription Orders`, `Promo-Codes`
Purpose	The core product definition form. Consolidates Plan Details, Pricing terms via a dynamic subform (cycles & currencies), and sets hard limits for quantitative resources and feature toggles.

2.1.2 Main Fields (Header Data Dictionary)

(Note: For quantitative limits, entering `-1` or checking an "Unlimited" box means No Hard Limit).

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`plan_id`	UUID (PK)	Yes	Auto	Hidden. Primary Key.
`plan_name_en`	VARCHAR(50)	Yes	None	Text Input (e.g., "Basic", "Pro").
`plan_name_ar`	VARCHAR(50)	Yes	None	Text Input (e.g., "الأساسية").
`base_cycle_days`	INT	Yes	30	Number Input. Fundamental duration of 1 cycle (e.g., 28, 30).
`is_active`	BOOLEAN	Yes	True	Toggle. If False, new tenants cannot subscribe to it.
`max_internal_users`	INT	Yes	None	Number Input. Staff users accessing the platform.
`max_internal_trucks`	INT	Yes	None	Number Input. Owned fleet size.
`max_external_trucks`	INT	Yes	None	Number Input. Subcontractor/Rented trucks.
`max_active_drivers`	INT	Yes	None	Number Input. Drivers registered and active.

`max_monthly_shipments`	INT	Yes	None	Number Input. Allowed Waybills per cycle.
`max_storage_gb`	INT	Yes	None	Number Input. Cloud attachment storage limit in GB.
`has_driver_app`	BOOLEAN	Yes	False	Toggle. Grants Tenant drivers access to the mobile app.
`backup_restore_level`	ENUM	Yes	'Standard'	Dropdown. Values: `Standard`, `Extended`, `Premium`.
`created_by`	UUID (FK)	Yes	Auto	Hidden. The Admin who created this plan.
`created_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.
`updated_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.

2.1.3 Subform 1: Plan Pricing Cycles

Behavior: A sub-table allowing multiple pricing rows for different cycle multipliers and currencies. Must be strictly linked to the Header.

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`pricing_id`	UUID (PK)	Yes	Auto	Hidden. Unique ID for this pricing line.
`plan_id`	UUID (FK)	Yes	None	Hidden. Linked strictly to the parent Plan Header.
`number_of_cycles`	INT	Yes	1	Number Input. The cycle multiplier (e.g., 1, 3, 12).
`currency_code`	FK	Yes	None	Dropdown. Fetched from Global `CurrencyMaster`.
`price`	DECIMAL(10,2)	Yes	0.00	Number Input. Total price for this specific duration and currency.

2.2 Global Add-ons Pricing Policy

2.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-02-02`
Form Name	Global Add-ons Pricing Policy
Form Type	Global Settings
Related Entities	`Subscription Orders` (Add-ons Subform)
Purpose	A fixed-field settings form dictating unit prices for exceeding plan limits. Ensures only ONE policy record is active at any time.

2.2.2 Main Fields (Data Dictionary)

(Pricing Assumed per 1 standard base cycle - System calculates per-day internally)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`policy_id`	UUID (PK)	Yes	Auto	Hidden. Primary Key.
`policy_name`	VARCHAR(50)	Yes	None	Text Input (e.g., "2026 Add-ons Rates").
`is_active`	BOOLEAN	Yes	False	Toggle. **System Rule:** True forces all other records to False.
`extra_internal_user_price`	DECIMAL(10,2)	Yes	0.00	Price for +1 Internal User per base cycle.
`extra_internal_truck_price`	DECIMAL(10,2)	Yes	0.00	Price for +1 Internal Truck per base cycle.
`extra_external_truck_price`	DECIMAL(10,2)	Yes	0.00	Price for +1 External Truck per base cycle.
`extra_driver_price`	DECIMAL(10,2)	Yes	0.00	Price for +1 Driver per base cycle.
`extra_shipment_price`	DECIMAL(10,2)	Yes	0.00	Price for +1 Shipment over the limit.
`extra_storage_gb_price`	DECIMAL(10,2)	Yes	0.00	Price for +1 GB Storage per base cycle.
`updated_by`	UUID (FK)	Yes	Auto	Hidden. The Admin who last modified the policy.
`updated_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.

2.3 Upgrade / Downgrade Policies (Backend Hardcoded Rules)

2.3.1 Overview

To ensure strict financial integrity, the logic for upgrading, downgrading, and Pro-Rata calculations is **hardcoded in the backend** as fixed business rules, rather than dynamic control panel settings.

2.3.2 The Fixed Business Logic

- A. Plan Upgrade Logic (Immediate Execution):
  1. System calculates the exact value of the unused days from the current plan (The Credit).
  2. System calculates the cost of the new plan for the chosen billing cycle.
  3. Tenant is immediately billed the difference: `(New Plan Cost) - (Unused Credit)`.
  4. Upon payment, new features are activated instantly, and a new billing cycle starts from that day.
- B. Plan Downgrade Logic (Deferred Execution):
  1. No refunds or credits are issued for downgrades.
  2. Tenant retains access to their current (higher) plan until the end of their current paid billing cycle.
  3. On the renewal date, the system automatically switches them to the lower plan and bills the lower amount. *(Must verify they meet lower plan limits first).*

- C. Add-ons Purchase Logic (Co-Terming):
  1. Calculates "Daily Rate" of the add-on `(Add-on Cycle Price / Base Cycle Days)`.
  2. Tenant is billed immediately for the remaining days of their current cycle: `(Daily Rate) * (Days Remaining)`.
  3. Add-on is activated instantly. At next renewal, base plan and full add-on price are merged into one invoice.

2.4 Promo-Codes Master (Offers & Discounts)

2.4.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-02-03`
Form Name	Promo-Codes Master
Form Type	Master File
Related Entities	`Subscription Orders`, `Subscription Plans Master`
Purpose	Allows generation of discount codes for marketing campaigns. Supports fixed/percentage discounts with strict expiry and usage limits.

2.4.2 Main Fields (Data Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`promo_code_id`	UUID (PK)	Yes	Auto	Hidden. Primary Key.
`code`	VARCHAR(20)	Yes	None	Text Input. **Strictly UNIQUE**, Alphanumeric, auto-UPPERCASE.
`discount_type`	ENUM	Yes	'Percent'	Dropdown. Values: `Percentage`, `Fixed_Amount`.
`discount_value`	DECIMAL(10,2)	Yes	None	Number Input. If `Percentage`, max is 100.
`discount_duration`	ENUM	Yes	'Once'	Dropdown. `Apply_Once` (First invoice only), `Recurring` (All renewals).
`valid_from`	TIMESTAMP	Yes	Current	Date/Time Picker. Code cannot be used before this.
`valid_until`	TIMESTAMP	No	Null	Date/Time Picker. Code expires after this.
`max_uses`	INT	No	Null	Number Input. Limits total times this code can be redeemed globally.
`current_uses`	INT	Yes	0	Read-only. Auto-increments upon successful payment.
`is_active`	BOOLEAN	Yes	True	Toggle. If False, code is instantly invalid regardless of dates.

`applicable_plans`	JSON (FK)	No	Null	Multi-select Dropdown (Plan IDs). If empty, applies to ALL plans.
`created_by`	UUID (FK)	Yes	Auto	Hidden. The Admin who created this promo code.
`created_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field.

2.4.3 Business Rules & Logic

- **Validation Order:** Before applying a code to an Order, the backend must sequentially check: `is_active` -> `valid_from/until` -> `max_uses > current_uses` -> `applicable_plans`. If any check fails, throw an "Invalid or Expired Code" error.

(End of CP-PCS-P4)

Control Panel PCS Part 5

Document ID: CP-PCS-P5 **Title:** Subscribers CRM & Billing **Phase:** 1 (MVP) **Module:** 4. Subscribers CRM & Billing

4.1 Subscriber Profile (Tenant Master Record)

4.1.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-04-01`
Form Name	Subscriber Profile (Tenant Master Record)
Form Type	Master File (Read/Write)
Related Entities	`Organization Profile` (Tenant DB), `Subscription Orders`, `Transactions`, `Standard Invoices`
Purpose	The central CRM record for a Tenant, combining synced organizational data, administrative controls, and the dynamically calculated active subscription state.

4.1.2 Main Fields (Data Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`tenant_id`	UUID (PK)	Yes	Auto	Hidden. Unique platform identifier.
`company_name`	VARCHAR(100)	Yes	None	Read-only Text. Synced from Tenant DB.
`registration_number`	VARCHAR(50)	Yes	None	Read-only Text. CR number. Synced from Tenant DB.
`tax_number`	VARCHAR(50)	Yes	None	Read-only Text. VAT/Tax Number. Synced from Tenant DB.
`primary_email`	VARCHAR(100)	Yes	None	Read-only Email. Synced from Tenant DB.
`primary_phone`	VARCHAR(20)	Yes	None	Read-only Phone. Synced from Tenant DB.
`country_code`	FK	Yes	None	Read-only. Synced from Global Country Master.
`account_status`	ENUM	Yes	'Active'	Dropdown. `Active`, `Suspended_Billing`, `Suspended_Violation`, `Churned`.
`assigned_sales_rep`	UUID (FK)	No	Null	Dropdown. Fetched from Admin Users.
`wallet_balance`	DECIMAL(10,2)	Yes	0.00	Read-only. Auto-calculated from Financial Ledger.

`registered_at`	TIMESTAMP	Yes	Auto	Read-only. Exact registration timestamp.
`total_ltv`	DECIMAL(10,2)	Yes	0.00	Read-only. Sum of all paid invoices (Lifetime Value).
`current_plan_id`	UUID (FK)	No	Null	Read-only Link. Active Base Plan (Auto-updated).
`subscription_start_date`	DATE	No	Null	Read-only. Date current cycle started (Auto-updated).
`subscription_expiry_date`	DATE	No	Null	Read-only. Date current cycle ends (Auto-updated).
`active_max_users`	INT	Yes	0	Read-only. Total users allowed (Auto-updated).
`active_max_internal_trucks`	INT	Yes	0	Read-only. Total owned trucks allowed (Auto-updated).
`active_max_external_trucks`	INT	Yes	0	Read-only. Total subcontracted trucks allowed (Auto-updated).
`active_max_drivers`	INT	Yes	0	Read-only. Total drivers allowed (Auto-updated).
`updated_at`	TIMESTAMP	Yes	Auto	System audit field.

4.1.3 Business Rules & Logic

- **The "Kill Switch":** Changing `account_status` to any `Suspended` state instantly revokes DB access for that Tenant schema, forcing logout for all associated users.
- **Data Synchronization:** Core profile changes made by the Tenant on their frontend automatically trigger a webhook to update the synced fields in this master record.
- **Owner Access (Impersonation):** "Login As" button generates a short-lived JWT to access the Tenant's workspace. Requires a permanent banner in UI and creates an audit log entry.

4.2 Internal CRM Notes

4.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-04-02`
Form Name	Internal CRM Notes
Form Type	Transactional (Sub-module of 4.1)
Related Entities	`Subscriber Profile`, `Admin Users`
Purpose	Internal timestamped logging for Sales/Support activities regarding the Tenant.

4.2.2 Main Fields (Data Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic

`note_id`	UUID (PK)	Yes	Auto	Hidden.
`tenant_id`	UUID (FK)	Yes	None	Hidden. Linked to Subscriber Profile.
`admin_id`	UUID (FK)	Yes	Auto	Hidden. The staff member who wrote the note.
`note_type`	ENUM	Yes	'General'	Dropdown. `General`, `Sales_Call`, `Billing_Issue`, `Complaint`.
`note_content`	TEXT	Yes	None	Text Area. The actual comment.
`created_at`	TIMESTAMP	Yes	Auto	System timestamp. Notes are strictly immutable.

4.3 Subscription Order Master

4.3.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-04-03`
Form Name	Subscription Order Master
Form Type	Transactional (Header-Detail)
Related Entities	`Subscription Plans`, `Promo-Codes`, `Transactions Ledger`, `Standard Invoices`, `Global Tax Settings`, `Payment Methods Configuration`, `IRoad FX Engine`
Purpose	Unified sales engine handling all subscription modifications with real-time projected state preview.

4.3.2 Main Fields (Header Data Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`order_id`	UUID (PK)	Yes	Auto	Hidden. Unique Order Number.
`tenant_id`	UUID (FK)	Yes	None	Hidden. Linked to Subscriber Profile.
`order_classification`	ENUM	Yes	'New'	Dropdown. `New_Subscription`, `Renewal`, `Upgrade`, `Downgrade`, `Add_ons`.
`promo_code_id`	UUID (FK)	No	Null	Dropdown/Text. Validates against active Promo-Codes.
`tax_id`	UUID (FK)	Yes	None	Hidden. Fetched from active `Global Tax Settings`.
`currency_code`	FK	Yes	None	Dropdown. Fetched from Global Currency Master based on Plan selection.
`sub_total`	DECIMAL(10,2)	Yes	0.00	Read-only. Sum of (Subform 1) + (Subform 2) in Transaction Currency.

`discount_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. Calculated dynamically if Promo Code applied.
`tax_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. Calculated based on `tax_id` percentage.
`grand_total`	DECIMAL(10,2)	Yes	0.00	Read-only. Final payable amount in Transaction Currency.
`exchange_rate_snapshot`	DECIMAL(10,6)	Yes	1.000000	Read-only. Snapshot of FX Rate from `IRoad FX Engine` (P9) at order creation.
`base_currency_equivalent`	DECIMAL(10,2)	Yes	0.00	Read-only. Computed: `grand_total` * `exchange_rate_snapshot`. (Used for pipeline projections).
`order_status`	ENUM	Yes	'Draft'	Dropdown. `Draft`, `Pending_Payment`, `Paid`, `Cancelled`.
`payment_method_id`	UUID (FK)	No	Null	Dropdown. Fetched dynamically from `Payment Methods Configuration` (P7).
`created_by`	UUID (FK)	Yes	Auto	Tracks if created by Sales Rep or auto-generated by Tenant.
`projected_plan_id`	UUID (FK)	No	Null	Read-only. Active Plan ID *after* payment (Preview).
`projected_expiry_date`	DATE	No	Null	Read-only. Recalculated expiry date (Preview).
`projected_max_users`	INT	No	Null	Read-only. Current Limit +/- Add-ons Subform (Preview).
`projected_max_internal_trucks`	INT	No	Null	Read-only. Current Limit +/- Add-ons Subform (Preview).
`projected_max_external_trucks`	INT	No	Null	Read-only. Current Limit +/- Add-ons Subform (Preview).
`projected_max_drivers`	INT	No	Null	Read-only. Current Limit +/- Add-ons Subform (Preview).
`created_at`	TIMESTAMP	Yes	Auto	System audit field.
`updated_at`	TIMESTAMP	Yes	Auto	System audit field.

4.3.3 Subform 1: Plan Selection

Behavior: Active only if classification involves a Plan. Strictly one row per order.

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`line_id`	UUID (PK)	Yes	Auto	Hidden. Unique record ID.
`order_id`	UUID (FK)	Yes	None	Hidden. Linked strictly to the Order Header.
`plan_id`	UUID (FK)	Yes	None	Dropdown. Fetched from `Subscription Plans`.
`number_of_cycles`	INT	Yes	1	Dropdown. Multiplier (e.g., 1, 3, 6, 12).

`plan_price`	DECIMAL(10,2)	Yes	0.00	Read-only. Fetched automatically.
`pro_rata_adjustment`	DECIMAL(10,2)	No	0.00	Read-only. Used for 'Upgrade' to deduct old plan value.
`line_total`	DECIMAL(10,2)	Yes	0.00	Read-only. `plan_price` + `pro_rata_adjustment`.

4.3.4 Subform 2: Add-ons Selection

Behavior: Handles extra resources. Co-terms with Base Plan automatically.

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`line_id`	UUID (PK)	Yes	Auto	Hidden. Unique record ID.
`order_id`	UUID (FK)	Yes	None	Hidden. Linked strictly to the Order Header.
`action_type`	ENUM	Yes	'Add'	Dropdown. `Add`, `Reduce`.
`add_on_type`	ENUM	Yes	None	Dropdown. `Extra_User`, `Extra_Internal_Truck`, etc.
`quantity`	INT	Yes	1	Number Input. The delta amount.
`number_of_cycles`	DECIMAL(10,2)	Yes	1.00	Read-only. Inherited from Plan OR Fraction of remaining days.
`unit_price`	DECIMAL(10,2)	Yes	0.00	Read-only. Fetched from `Add-ons Pricing Policy`.
`pro_rata_adjustment`	DECIMAL(10,2)	No	0.00	Read-only. Handles credits if reducing mid-cycle.
`line_total`	DECIMAL(10,2)	Yes	0.00	Read-only. If `Add`: `qty` * `unit_price` * `cycles`.

4.3.5 Business Rules & Logic

- **Provisioning Execution:** Tenant master limits and dates are **ONLY** overwritten when `order_status` == `Paid`.
- **Classification Logic:**
  - `New`: Activates immediately. Start Date = Today.
  - `Renewal`: Adds cycles to current expiry.
  - `Upgrade`: Immediate activation. Applies `pro_rata_adjustment` credit.
  - `Downgrade`: Takes effect only after current expiry date.
  - `Add_ons`: Immediate limit increase. Price pro-rated to current expiry date.

4.4 Financial Transactions Ledger

4.4.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-04-04`
Form Name	Financial Transactions Ledger
Form Type	Transactional / Financial Ledger
Related Entities	`Subscriber Profile`, `Subscription Order Master`, `Payment Methods Configuration`, `IRoad FX Engine`
Purpose	Source of truth for fund movements. Links to Orders or stands alone for manual wallet adjustments.

4.4.2 Main Fields (Data Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`transaction_id`	UUID (PK)	Yes	Auto	Hidden. Unique TXN Number.
`tenant_id`	UUID (FK)	Yes	None	Hidden. Transaction owner.
`order_id`	UUID (FK)	No	Null	Hidden. If null, this is a manual wallet adjustment.
`transaction_type`	ENUM	Yes	'Payment'	Dropdown. `Order_Payment`, `Wallet_TopUp`, `Refund`.
`payment_method_id`	UUID (FK)	Yes	None	Dropdown. Fetched dynamically from `Payment Methods Configuration` (P7).
`currency`	FK	Yes	None	Dropdown. Fetched from Global Currency Master.
`amount`	DECIMAL(10,2)	Yes	0.00	Number Input. The exact funds moved in transaction currency.
`exchange_rate_snapshot`	DECIMAL(10,6)	Yes	1.000000	Read-only. FX Rate from `IRoad FX Engine` (P9) at transaction time.
`base_currency_equivalent_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. Computed: `amount` * `exchange_rate_snapshot`.
`status`	ENUM	Yes	'Pending'	Dropdown. `Pending`, `Completed`, `Failed`, `Rejected`.
`gateway_ref`	VARCHAR(100)	No	Null	Text. External Stripe ID OR Bank Transfer receipt number.
`attachment_url`	VARCHAR(255)	No	Null	File Upload. **Mandatory** if `Offline_Bank` method is used.
`reviewed_by`	UUID (FK)	No	Null	Read-only. Admin user who approved/rejected.
`review_notes`	TEXT	No	Null	Text Area. Admin notes or rejection reason.
`created_at`	TIMESTAMP	Yes	Auto	Read-only. Exact time of the transaction creation.

`updated_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field (Records when status moves to Completed/Rejected).

4.4.3 Business Rules & Workflows

- **Automated Gateway Flow:** Created as `Pending`. Webhook updates to `Completed`. System auto-sets linked Order to `Paid`.
- **Bank Transfer Flow:** * If Approved by Admin: Status -> `Completed`. Sets Order to `Paid`.
  - If Rejected by Admin: Status -> `Rejected`. Requires `review_notes`. Order remains `Pending_Payment`.
- **Manual Wallet Top-Up:** Created directly by Admin (`order_id` is Null). Instantly adds credit to Tenant's `wallet_balance`.

4.5 Standard Invoice Record (نموذج الفاتورة المعياري)

4.5.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-04-05`
Form Name	Standard Invoice Record
Form Type	Transactional / Read-Only (System Generated)
Related Entities	`Subscription Order Master`, `Subscriber Profile`, `Global Tax Settings`, `IRoad FX Engine`
Purpose	A standardized, API-ready invoice container. Captures an immutable snapshot of transaction data, party details, and itemized financials.

4.5.2 Main Fields (Header & Parties Dictionary)

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`invoice_id`	UUID (PK)	Yes	Auto	Hidden. System ID.
`invoice_number`	VARCHAR(50)	Yes	Auto	Read-only. Sequential standard number (e.g., INV-2026-0001).
`order_id`	UUID (FK)	Yes	None	Read-only Link. The source Order of this invoice.
`tenant_id`	UUID (FK)	Yes	None	Read-only Link. The billed customer.
`tax_id`	UUID (FK)	Yes	None	Read-only Link. The active Tax Setting applied at the time of issue.
`issue_date`	TIMESTAMP	Yes	Auto	Read-only. The exact moment the invoice was generated.
`due_date`	DATE	Yes	Auto	Read-only. Usually matches `issue_date` for prepaid SaaS.
`status`	ENUM	Yes	'Issued'	Read-only. Values: `Draft`, `Issued`, `Paid`, `Void`.

`supplier_name`	VARCHAR(100)	Yes	None	Read-only. Hard-copied IRoad Company Name.
`supplier_tax_number`	VARCHAR(50)	Yes	None	Read-only. Hard-copied IRoad VAT Number.
`customer_name`	VARCHAR(100)	Yes	None	Read-only. Hard-copied Tenant `company_name` at time of issue.
`customer_tax_number`	VARCHAR(50)	No	Null	Read-only. Hard-copied Tenant `tax_number` at time of issue.
`customer_address`	TEXT	No	Null	Read-only. Hard-copied Tenant address at time of issue.
`sub_total`	DECIMAL(10,2)	Yes	0.00	Read-only. Total before tax and discount in transaction currency.
`discount_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. Applied discount from promo code.
`taxable_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. `sub_total` - `discount_amount`.
`tax_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. Total VAT amount based on snapshot tax rate.
`grand_total`	DECIMAL(10,2)	Yes	0.00	Read-only. Final amount strictly matching the Transaction.
`currency_code`	FK	Yes	None	Read-only. Copied from the transaction/order.
`exchange_rate_snapshot`	DECIMAL(10,6)	Yes	1.000000	Read-only. Snapshot of FX Rate (P9) at time of issue.
`base_currency_equivalent_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. Computed: `grand_total` * `exchange_rate_snapshot`.
`updated_at`	TIMESTAMP	Yes	Auto	Read-only. System audit field (Useful if status changes to Void).

4.5.3 Subform 1: Invoice Line Items

Behavior: Itemized translation of the Order Subforms into a strict accounting format.

Field Name	Data Type	Req	Default	UI Element & Validation / Logic
`line_id`	UUID (PK)	Yes	Auto	Hidden.
`invoice_id`	UUID (FK)	Yes	None	Hidden. Linked strictly to the Invoice Header.
`item_description`	VARCHAR(255)	Yes	None	Read-only. Descriptive text (e.g., "Basic Plan - Annual", "Add-on: 5 Users").
`quantity`	DECIMAL(10,2)	Yes	1.00	Read-only. Quantity of the item.
`unit_price`	DECIMAL(10,2)	Yes	0.00	Read-only. Price per unit before tax.
`tax_rate`	DECIMAL(5,2)	Yes	0.00	Read-only. **Snapshot** of the percentage from `tax_id` (e.g., 15.00).

`tax_amount`	DECIMAL(10,2)	Yes	0.00	Read-only. Calculated tax for this specific line.
`line_total`	DECIMAL(10,2)	Yes	0.00	Read-only. `(quantity * unit_price) + tax_amount`.

4.5.4 Business Rules & Logic

- **Generation Trigger:** The system automatically generates and locks this Standard Invoice Record exactly when an Order's status transitions to `Paid`.
- **Absolute Immutability (Snapshotting):** Once generated, NO field in the Invoice or its Line Items can be modified by any user or admin. Supplier, Customer details, and Tax Rates are hard-copied (snapshot) so that future changes to the Tenant's profile or Global Tax Settings do not alter historical legal documents.
- **Voiding / Reversal:** If an invoice contains an error or a refund is issued, the original invoice is never deleted. Its status may change to `Void`, and a separate "Credit Note" (or negative invoice) must be generated to satisfy accounting principles.

(End of CP-PCS-P5)

Control Panel PCS Part 6

Document ID: CP-PCS-P6 **Title:** Communication & Alerts **Phase:** 1 (MVP) **Module:** 5. Communication & Alerts

5.1 Communication Gateways Settings (SMTP & SMS)

5.1.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-05-01`
Form Name	Communication Gateways Settings
Form Type	System Settings (Read/Write)
Related Entities	`Events Mapping Engine`, `Communication Logs`
Purpose	Centralized configuration for external communication providers (Email SMTP and SMS APIs). Prevents hardcoding credentials in the backend and allows instant switching of providers with zero downtime.

5.1.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`gateway_id`	Gateway ID	معرف البوابة	STORED	Text	Yes	Auto-number	Primary Key
2	`gateway_type`	Gateway Type	نوع البوابة	STORED	Enum	Yes	Enum: [Email, SMS]	Must be one of enum values
3	`provider_name`	Provider Name	اسم المزود	STORED	Text	Yes	-	e.g., "SendGrid", "Unifonic"
4	`host_url`	Host / API URL	الرابط / الخادم	STORED	Text	Yes	-	SMTP Host or SMS API Base URL
5	`port`	Port	المنفذ	STORED	Integer	No	-	Primarily for SMTP (e.g., 587, 465)
6	`username_key`	Username / Key	اسم المستخدم / المفتاح	STORED	Text	Yes	-	SMTP Username or SMS API Key
7	`password_secret`	Password / Secret	كلمة المرور / السر	STORED	Text	Yes	-	Masked input
8	`sender_id`	Sender ID	معرف المرسل	STORED	Text	Yes	-	From Email Address OR SMS Sender ID
9	`encryption_type`	Encryption Type	نوع التشفير	STORED	Enum	No	Enum: [TLS, SSL, None]	Mainly for SMTP
10	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Default: False

11	`updated_by`	Updated By	تم التحديث بواسطة	STORED	FK	Yes	UserAccount	Audit field

5.1.3 Business Rules & Logic

- **Singularity Rule:** The system can store multiple gateways, but only **ONE** gateway per `gateway_type` can have `is_active = True` at any given time. Activating one automatically deactivates the others of the same type.
- **Test Connection:** UI must include a "Test Connection" button that attempts to send a dummy message using the entered credentials before saving.

5.2 Notification Templates Master (Email & SMS)

5.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-05-02`
Form Name	Notification Templates Master
Form Type	Master File (Read/Write)
Related Entities	`Events Mapping Engine`
Purpose	Stores the multilingual, dynamic content for **Email and SMS** messages. Separates the message body from the backend code.

5.2.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`template_id`	Template ID	معرف القالب	STORED	Text	Yes	Auto-number	Primary Key
2	`template_name`	Template Name	اسم القالب	STORED	Text	Yes	-	Internal identifier (e.g., "Welcome Email - V1")
3	`channel_type`	Channel Type	نوع القناة	STORED	Enum	Yes	Enum: [Email, SMS]	Determines editor type
4	`category`	Category	التصنيف	STORED	Enum	Yes	Enum: [Transactional, Promotional]	Transactional is mandatory (e.g., OTP)
5	`subject_en`	Subject (EN)	العنوان (إنجليزي)	STORED	Text	Cond	-	Required if `channel_type` = Email
6	`subject_ar`	Subject (AR)	العنوان (عربي)	STORED	Text	Cond	-	Required if `channel_type` = Email
7	`body_en`	Body (EN)	المحتوى (إنجليزي)	STORED	Text	Yes	-	Supports `{{variables}}`
8	`body_ar`	Body (AR)	المحتوى (عربي)	STORED	Text	Yes	-	Supports `{{variables}}`
9	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Default: True

10	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount	Audit field

5.2.3 Business Rules & Logic

- **Dynamic Shortcodes:** The backend provides a standardized list of variables (e.g., `{{client_name}}`, `{{otp_code}}`, `{{invoice_amount}}`) that can be injected into the `body` fields. The system dynamically replaces these before dispatching.

5.3 System Events Mapping Engine

5.3.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-05-03`
Form Name	System Events Mapping Engine
Form Type	Configuration / Rules Engine
Related Entities	`Notification Templates`, `Gateways`
Purpose	The intelligent router. Links hardcoded backend system events (Triggers) to the dynamically created Notification Templates (Email/SMS). Configures Fallback mechanisms for critical events like OTP.

5.3.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`mapping_id`	Mapping ID	معرف الربط	STORED	Text	Yes	Auto-number	Primary Key
2	`system_event`	System Event	حدث النظام	STORED	Enum	Yes	**Fixed Backend List**	e.g., `OTP_Requested`, `Password_Changed`, `Invoice_Paid`
3	`primary_channel`	Primary Channel	القناة الأساسية	STORED	Enum	Yes	Enum: [Email, SMS]	Defines the main delivery method
4	`primary_template_id`	Primary Template	القالب الأساسي	STORED	FK	Yes	Notification Templates	Filtered by `primary_channel`
5	`fallback_channel`	Fallback Channel	القناة البديلة	STORED	Enum	No	Enum: [Email, SMS]	Executed if primary fails
6	`fallback_template_id`	Fallback Template	القالب البديل	STORED	FK	Cond	Notification Templates	Required if `fallback_channel` is selected
7	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Default: True
8	`updated_by`	Updated By	المعدل	STORED	FK	Yes	UserAccount	Audit field

5.3.3 Business Rules & Logic

- **Event Immutability Rule (Architectural Core):** The list of `system_event` triggers is **strictly hardcoded** by the backend development team (e.g., `PASSWORD_CHANGED`). Super Admins **cannot** create new event types dynamically from the UI. Their role is strictly to select an existing event from the dropdown and map it to a template.
- **The Fallback Protocol:** If a system event fires and the `primary_channel` (e.g., SMS for OTP) returns a "Failure" or "Timeout" from the Gateway API, the system instantly catches the exception and dispatches via the `fallback_channel` (e.g., Email) using the `fallback_template_id`.

5.4 Push Notifications Manager

5.4.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-05-04`
Form Name	Push Notifications Manager
Form Type	Transactional / Configuration
Purpose	A unified interface to create Push Notification content and define its execution logic (either as a Manual Broadcast to a specific audience, or linked dynamically to a fixed System Event).

5.4.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`notification_id`	Notification ID	معرف الإشعار	STORED	Text	Yes	Auto-number	Primary Key
2	`internal_name`	Internal Name	الاسم المرجعي	STORED	Text	Yes	-	e.g., "Driver Welcome", "Eid Promo"
3	`title_en`	Title (EN)	العنوان (إنجليزي)	STORED	Text	Yes	-	Push Notification Title
4	`title_ar`	Title (AR)	العنوان (عربي)	STORED	Text	Yes	-	Push Notification Title
5	`message_en`	Message (EN)	الرسالة (إنجليزي)	STORED	Text	Yes	-	Supports `{{variables}}` for events
6	`message_ar`	Message (AR)	الرسالة (عربي)	STORED	Text	Yes	-	Supports `{{variables}}` for events
7	`action_link`	Action Link	رابط الإجراء	STORED	URL	No	-	Deep Link inside App
8	`trigger_mode`	Trigger Mode	نوع الإطلاق	STORED	Enum	Yes	Enum: [Manual_Broadcast, System_Event]	Defines execution behavior
9	`linked_event`	Linked Event	الحدث المرتبط	STORED	Enum	Cond	**Fixed Backend List**	Required if `trigger_mode` = System_Event
10	`target_audience`	Target Audience	الجمهور المستهدف	STORED	Enum	Cond	[All, Tenants, Drivers, Specific]	Required if `trigger_mode` = Manual_Broadcast
11	`specific_target_id`	Specific Target	الهدف المخصص	STORED	FK	Cond	ClientAccount / Driver	Required if Target Audience = Specific

12	`scheduled_at`	Scheduled At	موعد الإرسال	STORED	DateTime	No	-	For Manual_Broadcast only
13	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	For System_Event only
14	`dispatch_status`	Status	حالة الإرسال	STORED	Enum	Yes	[Draft, Scheduled, Completed]	For Manual_Broadcast only
15	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount	Audit field

5.4.3 Business Rules & Logic

- **Event Selection Rule:** Just like Email/SMS mapping, `linked_event` relies on a fixed, hardcoded list programmed in the backend (e.g., `Shipment_Assigned_to_Driver`).
- **Execution Logic:**
  - **System Event (`trigger_mode` = System_Event):** The record acts as an active rule. Scheduling (`scheduled_at`) and Dispatch Status (`dispatch_status`) are irrelevant and hidden. The push fires whenever the backend triggers the event.
  - **Manual Broadcast (`trigger_mode` = Manual_Broadcast):** The record acts as a one-time campaign. It utilizes a background queue (e.g., Redis) to push to FCM tokens without blocking the UI. `linked_event` is irrelevant and hidden.

5.5 System Banners (In-App Announcements)

5.5.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-05-05`
Form Name	System Banners
Form Type	Master File
Purpose	Manages non-intrusive global announcements displayed at the top of the Tenant Control Panel (e.g., Scheduled Maintenance).

5.5.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`banner_id`	Banner ID	معرف البانر	STORED	Text	Yes	Auto-number	Primary Key
2	`title_en`	Title (EN)	العنوان (إنجليزي)	STORED	Text	Yes	-	-
3	`title_ar`	Title (AR)	العنوان (عربي)	STORED	Text	Yes	-	-
4	`message_en`	Message (EN)	الرسالة (إنجليزي)	STORED	Text	Yes	-	-
5	`message_ar`	Message (AR)	الرسالة (عربي)	STORED	Text	Yes	-	-

6	`severity`	Severity	الأهمية (اللون)	STORED	Enum	Yes	Enum: [Info, Warning, Critical]	Determines UI color (Blue, Yellow, Red)
7	`is_dismissible`	Dismissible?	قابل للإغلاق؟	STORED	Boolean	Yes	-	If False, the X button is hidden
8	`valid_from`	Valid From	يظهر من	STORED	DateTime	Yes	-	Current time or future
9	`valid_until`	Valid Until	يختفي في	STORED	DateTime	No	-	Auto-hides after this time
10	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Manual override switch

5.6 Internal Alerts Routing

5.6.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-05-06`
Form Name	Internal Alerts Routing
Form Type	Configuration
Purpose	Automates internal notifications so IRoad staff (Sales/Support) are instantly informed of critical tenant activities.

5.6.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`route_id`	Route ID	معرف التوجيه	STORED	Text	Yes	Auto-number	Primary Key
2	`trigger_event`	Trigger Event	الحدث	STORED	Enum	Yes	**Fixed Backend List**	e.g., `New_Tenant_Registered`, `High_Priority_Ticket`
3	`notify_role_id`	Notify Role	إشعار فريق	STORED	FK	No	UserRole	Sends to all Admins with this Role
4	`notify_custom_email`	Custom Email	إيميل مخصص	STORED	Text	No	-	e.g., `billing-alerts@iroad.com`
5	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Default: True

5.7 Communication & Delivery Logs (Audit Trail)

5.7.1 Form Identification Card

Attribute	Description

Form ID	`FRM-CP-05-07`
Form Name	Communication Logs
Form Type	Audit / Read-Only
Purpose	The ultimate truth for all outgoing messages. Essential for Support Agents to verify if a tenant actually received an OTP or Invoice.

5.7.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`log_id`	Log ID	السجل	STORED	Text	Yes	Auto-number	Primary Key
2	`recipient`	Recipient	المستلم	STORED	Text	Yes	-	Email Address, Phone Number, or FCM Token
3	`client_id`	Client Ref	العميل المرتبط	STORED	FK	No	ClientAccount	For fast filtering
4	`channel_type`	Channel Type	نوع القناة	STORED	Enum	Yes	Enum: [Email, SMS, Push]	-
5	`trigger_source`	Trigger Source	المصدر	STORED	Text	Yes	-	e.g., "Event: OTP_Requested" or "Campaign: V1"
6	`delivery_status`	Delivery Status	حالة الإرسال	STORED	Enum	Yes	[Sent, Failed, Bounced]	Based on Gateway API response
7	`error_details`	Error Details	تفاصيل الخطأ	STORED	Text	No	-	Exact API error (e.g., "Invalid Number")
8	`dispatched_at`	Dispatched At	وقت الإرسال	STORED	DateTime	Yes	System	Exact execution time

5.7.3 Business Rules & Logic

- **Immutability:** This log is strictly append-only. No Super Admin or Database Administrator can edit or delete a log entry.
- **Retention Policy:** To prevent DB bloat, logs older than 90 days are automatically archived to cold storage, keeping only the recent transactional history readily available in the UI.

(End of CP-PCS-P6)

Control Panel PCS Part 7

Document ID: CP-PCS-P7 **Title:** System Settings & Payment Infrastructure **Phase:** 1 (MVP) **Module:** 7. System Settings

7.1 IRoad Bank Accounts Master

7.1.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-07-01`
Form Name	IRoad Bank Accounts Master
Form Type	System Settings (Master Data)
Related Entities	`Payment Methods Configuration`
Purpose	Stores the official bank accounts of IRoad. These accounts are displayed to Tenants during the checkout process when they select "Bank Transfer" (Offline) as their payment method.

7.1.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`account_id`	Account ID	معرف الحساب	STORED	Text	Yes	Auto-number	Primary Key
2	`bank_name`	Bank Name	اسم البنك	STORED	Text	Yes	-	e.g., "Al Rajhi Bank", "SNB"
3	`account_holder_name`	Account Holder Name	اسم صاحب الحساب	STORED	Text	Yes	-	e.g., "IRoad Technology Co."
4	`iban_number`	IBAN Number	رقم الأيبان	STORED	Text	Yes	-	Must pass standard IBAN format validation
5	`account_number`	Account Number	رقم الحساب	STORED	Text	Yes	-	Numeric only
6	`swift_code`	SWIFT Code	سويفت كود	STORED	Text	No	-	Standard Bank Identifier Code
7	`currency_code`	Currency	عملة الحساب	STORED	FK	Yes	CurrencyMaster.currency_code	Defines which invoices can be paid into this account
8	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Default: True. If False, hidden from Tenants.
9	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount.id	Audit field
10	`created_at`	Created At	تاريخ الإنشاء	STORED	DateTime	Yes	System	Audit field

11	`updated_by`	Updated By	المعدل	STORED	FK	No	UserAccount.id	Audit field
12	`updated_at`	Updated At	آخر تحديث	STORED	DateTime	Yes	System	Audit field

7.1.3 Business Rules & Logic

- **Dynamic Display & Currency Matching:** The frontend checkout page will iterate over records where `is_active = True` AND `currency_code` matches the current Order/Invoice currency, displaying them as destination accounts for manual bank transfers.

7.2 Payment Gateways API Settings

7.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-07-02`
Form Name	Payment Gateways API Settings
Form Type	System Settings (Security/Integration)
Related Entities	`Payment Methods Configuration`
Purpose	Secure storage for third-party payment provider credentials. Uses a dynamic JSON payload to abstract varying key requirements across different providers (Stripe vs PayTabs vs Moyasar).

7.2.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`gateway_id`	Gateway ID	معرف البوابة	STORED	Text	Yes	Auto-number	Primary Key
2	`gateway_name`	Gateway Name	اسم البوابة	STORED	Text	Yes	-	e.g., "Moyasar", "Stripe", "PayTabs"
3	`environment`	Environment	بيئة العمل	STORED	Enum	Yes	[Test, Live]	Determines which keys are active
4	`credentials_payload`	Credentials (JSON)	بيانات الربط	STORED	JSON	Yes	-	Masked. Dynamic object holding all necessary keys (e.g., `{"public_key": "...", "secret_key": "...", "profile_id": "..."}`)
5	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Master switch for the entire gateway integration
6	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount.id	Audit field
7	`created_at`	Created At	تاريخ الإنشاء	STORED	DateTime	Yes	System	Audit field
8	`updated_by`	Updated By	المعدل	STORED	FK	No	UserAccount.id	Audit field

9	`updated_at`	Updated At	آخر تحديث	STORED	DateTime	Yes	System	Audit field

7.2.3 Business Rules & Logic

- **Dynamic Credentials:** By using a JSON `credentials_payload`, the system future-proofs itself. The backend gateway drivers will extract the specific keys they need based on the `gateway_name` without requiring database schema changes.

7.3 Payment Methods Configuration (Dynamic Engine)

7.3.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-07-03`
Form Name	Payment Methods Configuration
Form Type	Rules Engine / Master File
Related Entities	`Payment Gateways API`, `IRoad Bank Accounts`, `Subscription Orders` (CP-PCS-P5)
Purpose	The dynamic bridge between backend settings and the Tenant's frontend checkout. Defines payment options, currencies supported, and processing flow (Online vs Offline).

7.3.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`method_id`	Method ID	معرف الطريقة	STORED	Text	Yes	Auto-number	Primary Key
2	`method_name_en`	Method Name (EN)	اسم الطريقة (إنجليزي)	STORED	Text	Yes	-	Displayed to Tenant (e.g., "Credit Card", "Apple Pay", "Bank Transfer USD")
3	`method_name_ar`	Method Name (AR)	اسم الطريقة (عربي)	STORED	Text	Yes	-	Displayed to Tenant (e.g., "بطاقة ائتمانية", "حوالة بنكية دولية")
4	`method_type`	Method Type	نوع المعالجة	STORED	Enum	Yes	[Online_Gateway, Offline_Bank]	Determines the checkout flow logic
5	`supported_currencies`	Supported Currencies	العملات المدعومة	STORED	JSON	Yes	CurrencyMaster.code	Array of currencies this method accepts (e.g., `["SAR", "USD"]`).
6	`gateway_id`	Associated Gateway	البوابة المرتبطة	STORED	FK	Cond	Payment Gateways API	**Required if** `method_type` = Online_Gateway
7	`dedicated_bank_account_id`	Dedicated Bank Account	الحساب البنكي المخصص	STORED	FK	No	IRoad Bank Accounts	Optional override. Directs funds to a specific bank account if `method_type` = Offline_Bank.
8	`logo_url`	Logo Icon	أيقونة الطريقة	STORED	File	No	-	Visual identifier (e.g., Visa/Mada icon)

9	`display_order`	Display Order	ترتيب العرض	STORED	Integer	Yes	-	Controls sorting in the checkout UI
10	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	If False, option disappears from checkout instantly
11	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount.id	Audit field
12	`created_at`	Created At	تاريخ الإنشاء	STORED	DateTime	Yes	System	Audit field
13	`updated_by`	Updated By	المعدل	STORED	FK	No	UserAccount.id	Audit field
14	`updated_at`	Updated At	آخر تحديث	STORED	DateTime	Yes	System	Audit field

7.3.3 Business Rules & Logic

- The Checkout Router Logic (Currency & Type Aware):
  - When rendering the Tenant Checkout page, the system fetches all records from `FRM-CP-07-03` where `is_active = True` AND the Order's currency is present in the `supported_currencies` array.
  - If `method_type = Offline_Bank`:
    - If `dedicated_bank_account_id` is set, the system displays **ONLY** that specific bank account to the user.
    - If `dedicated_bank_account_id` is NULL, the system fetches all active `IRoad Bank Accounts` (FRM-CP-07-01) that match the Order's currency and displays them.
    - Prompts the user to upload a transfer receipt. The Order status becomes `Pending_Payment`.
  - If `method_type = Online_Gateway`:
    - The system fetches the dynamic credentials from the associated `gateway_id` (FRM-CP-07-02) and initializes the digital payment session. Upon successful webhook verification, the Order status automatically becomes `Paid`.

(End of CP-PCS-P7)

Control Panel PCS Part 8

Document ID: CP-PCS-P8 **Title:** Global Master Data Management **Phase:** 1 (MVP) **Module:** 8. Global Master Data

8.1 Countries Master (إدارة الدول)

8.1.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-08-01`
Form Name	Countries Master
Form Type	System Settings (Master Data)
Original Reference	Country Master 8 صفحة
Purpose	Centralized management of supported countries. This data feeds into Tenant registration, addresses, and localization settings across the platform.

8.1.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`country_code`	Country Code	كود الدولة	STORED	Text	Yes	-	**Primary Key**. Must be formatted as ISO Country Code (Unique).
2	`name_en`	English Name	اسم الدولة بالإنجليزي	STORED	Text	Yes	-	Text input.
3	`name_ar`	Arabic Name	اسم الدولة بالعربي	STORED	Text	Yes	-	Text input.
4	`is_active`	Status	الحالة	STORED	Boolean	Yes	-	Default: True. Only active countries appear in registration/dropdowns.
5	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount	System audit field.
6	`updated_at`	Updated At	آخر تحديث	STORED	DateTime	Yes	System	System audit field.

8.1.3 Business Rules & Logic

- **No Hard Delete:** Deleting a country that is linked to existing Tenants will cause database integrity issues. Therefore, countries can only be deactivated (`is_active = False`), not deleted.
- **Global Caching:** To reduce database load during Tenant registration or checkout, the list of active countries must be cached (e.g., via Redis) and only refreshed when a Super Admin modifies this form.

8.2 Currencies Master (إدارة العملات)

8.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-08-02`
Form Name	Currencies Master
Form Type	System Settings (Master Data)
Original Reference	CUR-001 page 40
Purpose	Core financial configuration module defining all accepted currencies. Directly impacts Subscription Plans, Payment Gateways, and Invoicing.

8.2.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`currency_code`	Currency Code	كود العملة	STORED	Text	Yes	-	**Primary Key**. Must be formatted as ISO Code (e.g., SAR, USD) and strictly **Unique**.
2	`name_en`	Currency Name - English	اسم العملة بالإنجليزي	STORED	Text	Yes	-	Text input.
3	`name_ar`	Currency Name - Arabic	اسم العملة بالعربي	STORED	Text	Yes	-	Text input.
4	`currency_symbol`	Currency Symbol	رمز العملة	STORED	Text	Yes	-	Visual symbol (e.g., $, £, ريال).
5	`decimal_places`	Decimal Places	الخانات العشرية	STORED	Integer	No	-	Optional. Defines precision for rounding (Recommended: 2).
6	`is_active`	Status	الحالة	STORED	Boolean	Yes	-	Active/Inactive. **Only active currencies can be selected by Tenants.**
7	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount	System audit field.
8	`updated_at`	Updated At	آخر تحديث	STORED	DateTime	Yes	System	System audit field.

8.2.3 Business Rules & Logic

- **Tenant Visibility Rule:** As specified in the original constraints, only currencies marked with status `Active` will be available for Tenants to select or use within the system.
- **Immutability of Code:** Once a `currency_code` (e.g., SAR) is saved and linked to transactions or plans, it becomes immutable (cannot be edited) to prevent historical financial data corruption.
- **System Settings Integration:** This table acts as the lookup source (`Foreign Key`) for the `supported_currencies` field defined in the Payment Methods Configuration (`FRM-CP-07-03`).

(End of CP-PCS-P8)

Control Panel PCS Part 9

Document ID: CP-PCS-P9 **Title:** System Configurations & Tax Settings **Phase:** 1 (MVP) **Module:** 9. System Configurations

9.1 Tax Codes Master (نظام أكواد الضرائب الذكي)

9.1.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-09-01`
Form Name	Tax Codes Master
Form Type	Master File (Configuration)
Related Entities	`Countries Master` (P8), `Standard Invoice Record` (P5)
Purpose	Defines specific tax rules and percentages applied by IRoad. Features dynamic routing based on the Tenant's physical/IP location.

9.1.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`tax_code`	Tax Code	كود الضريبة	STORED	Text	Yes	-	**Primary Key**. Unique (e.g., S-15, Z-0).
2	`name_en`	Name (English)	الاسم الإنجليزي	STORED	Text	Yes	-	e.g., "Standard VAT 15%"
3	`name_ar`	Name (Arabic)	الاسم العربي	STORED	Text	Yes	-	e.g., "ضريبة القيمة المضافة 15%"
4	`rate_percent`	Rate %	النسبة	STORED	Decimal	Yes	-	0.00 to 100.00
5	`applicable_country_code`	Applicable Country	دولة التطبيق	STORED	FK	No	Countries Master	Links this tax to a specific country (e.g., SA).
6	`is_default_for_country`	Default for Country?	افتراضي لهذه الدولة؟	STORED	Boolean	Yes	-	Prevents ambiguity. Only ONE code can be default per country.
7	`is_international_default`	Int. Default?	افتراضي دولي؟	STORED	Boolean	Yes	-	Applies to any country not mapped. Only ONE can be True globally.
8	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Default: True.
9	`created_by`	Created By	المنشئ	STORED	FK	Yes	UserAccount.id	Audit field
10	`created_at`	Created At	تاريخ الإنشاء	STORED	DateTime	Yes	System	Audit field

11	`updated_by`	Updated By	المعدل	STORED	FK	No	UserAccount.id	Audit field
12	`updated_at`	Updated At	آخر تحديث	STORED	DateTime	Yes	System	Audit field

9.1.3 Business Rules & Logic

- **Dynamic Routing Rule:** When generating an invoice, the system checks the Tenant's country. It queries this table for an active tax code where `applicable_country_code == Tenant.Country` AND `is_default_for_country == True`.
- **Fallback Logic:** If no specific default tax code exists for that country, it falls back to the tax code where `is_international_default == True`.

9.2 IRoad Billing & Tax Settings (إعدادات الفوترة والتحقق)

9.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-09-02`
Form Name	General Tax Settings
Form Type	Configuration (Single Record)
Purpose	Sets the default financial behavior and defines how the system verifies the Tenant's location to prevent tax evasion without blocking checkouts.

9.2.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`setting_id`	Setting ID	المعرف	STORED	Text	Yes	-	Fixed ID (e.g., "GLOBAL-TAX-SETTING")
2	`prices_include_tax`	Prices Include Tax?	الأسعار شاملة الضريبة؟	STORED	Boolean	Yes	-	Default: False (B2B standard logic)
3	`location_verification`	Location Verification	التحقق من الموقع	STORED	Enum	Yes	$Profile_Only, Audit_Only, Enforce_Profile_Match$	Defines anti-evasion logic.
4	`updated_by`	Updated By	آخر تحديث بواسطة	STORED	FK	Yes	UserAccount.id	Audit field
5	`updated_at`	Updated At	وقت التحديث	STORED	DateTime	Yes	System	Audit field

9.2.3 Business Rules & Logic

- **Location Verification Logic:**
  - `Profile_Only`: Relies solely on the Country selected by the Tenant during registration.
  - `Audit_Only`: System captures IP Geolocation during checkout. If it mismatches the profile country, it logs the IP in the invoice for auditing but **allows** the payment to proceed based on profile country.

- `Enforce_Profile_Match` : If IP Geolocation mismatches the profile country, the checkout is **blocked**, and the Tenant is prompted to update their profile address to match their current physical location to proceed.

9.3 IRoad Legal Identity (الهوية القانونية لشركة مسار)

9.3.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-09-03`
Form Name	IRoad Legal Identity
Form Type	Configuration (Single Record)
Related Entities	`Standard Invoice Record` (P5), `Countries Master` (P8)
Purpose	Stores the official corporate details, Point of Origin, and visual identity (Logo) of the platform operator for legal invoice generation.

9.3.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`identity_id`	Identity ID	المعرف	STORED	Text	Yes	-	Fixed ID (e.g., "GLOBAL-LEGAL-IDENTITY")
2	`company_logo`	Company Logo	شعار الشركة	STORED	File	Yes	-	Image file URL. Placed at the header of PDF invoices.
3	`company_name_en`	Company Name (EN)	اسم الشركة (إنجليزي)	STORED	Text	Yes	-	e.g., "IRoad Technology"
4	`company_name_ar`	Company Name (AR)	اسم الشركة (عربي)	STORED	Text	Yes	-	e.g., "شركة مسار لتقنية المعلومات"
5	`company_country_code`	HQ Country (Origin)	دولة المقر (المنشأ)	STORED	FK	Yes	Countries Master	**Crucial:** Defines if a sale is Domestic or Cross-border Export.
6	`commercial_register`	Commercial Register	السجل التجاري	STORED	Text	Yes	-	Numeric format
7	`tax_number`	VAT/Tax Number	الرقم الضريبي	STORED	Text	Yes	-	Standard VAT length
8	`registered_address`	Registered Address	العنوان الوطني	STORED	Text	Yes	-	Complete address string
9	`support_email`	Support Email	بريد الدعم الفني	STORED	Text	Yes	-	Valid email format
10	`support_phone`	Support Phone	هاتف الدعم الفني	STORED	Text	No	-	e.g., "9200..."
11	`updated_by`	Updated By	آخر تحديث بواسطة	STORED	FK	Yes	UserAccount.id	Audit field
12	`updated_at`	Updated At	وقت التحديث	STORED	DateTime	Yes	System	Audit field

9.4 Global System & Subscription Rules (قواعد النظام والاشتراكات)

9.4.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-09-04`
Form Name	Global System Rules
Form Type	Configuration (Single Record)
Purpose	Defines the core logic parameters that govern the SaaS mechanics, including time zones and grace periods.

9.4.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`rule_id`	Rule ID	المعرف	STORED	Text	Yes	-	Fixed ID (e.g., "GLOBAL-SYSTEM-RULES")
2	`system_timezone`	System Timezone	المنطقة الزمنية	STORED	Text	Yes	-	e.g., "Asia/Riyadh". Crucial for Expiry Cron Jobs.
3	`default_date_format`	Default Date Format	صيغة التاريخ	STORED	Enum	Yes	$YYYY-MM-DD, DD/MM/YYYY$	Determines display format in CP
4	`grace_period_days`	Grace Period (Days)	فترة السماح (أيام)	STORED	Integer	Yes	-	Days a tenant can operate post-expiry before auto-suspension.
5	`standard_billing_cycle`	Billing Cycle (Days)	طول دورة الفوترة	STORED	Integer	Yes	-	Default: 30. Base multiplier for Pro-Rata calculations.
6	`updated_by`	Updated By	آخر تحديث بواسطة	STORED	FK	Yes	UserAccount.id	Audit field
7	`updated_at`	Updated At	وقت التحديث	STORED	DateTime	Yes	System	Audit field

9.5 IRoad Base Currency Configuration (عملة الأساس للآدمن)

9.5.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-09-05`
Form Name	IRoad Base Currency
Form Type	Configuration (Single Record)

Purpose	Defines the core accounting currency for IRoad platform. All financial dashboard reports and LTV calculations will be converted and displayed in this currency.

9.5.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`setting_id`	Setting ID	المعرف	STORED	Text	Yes	-	Fixed ID (e.g., "GLOBAL-BASE-CURRENCY")
2	`base_currency_id`	Base Currency	عملة الأساس للتقارير	STORED	FK	Yes	Currencies Master (P8)	**Immutable** after initialization/Phase 1.
3	`updated_by`	Updated By	آخر تحديث بواسطة	STORED	FK	Yes	UserAccount.id	Audit field
4	`updated_at`	Updated At	وقت التحديث	STORED	DateTime	Yes	System	Audit field

9.5.3 Business Rules & Logic

- **Immutability Rule:** Once set and the first financial transaction is recorded, `base_currency_id` becomes strictly locked (Read-only) to prevent catastrophic corruption of historical financial reporting.

9.6 IRoad Exchange Rates Manager (إدارة أسعار الصرف)

9.6.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-09-06`
Form Name	IRoad Exchange Rates
Form Type	Master File (Configuration)
Purpose	Manages the Foreign Exchange (FX) rates for all accepted secondary currencies against the IRoad Base Currency.

9.6.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`fx_id`	FX ID	معرف الصرف	STORED	Text	Yes	Auto-number	Primary Key
2	`currency_id`	Secondary Currency	العملة المستهدفة	STORED	FK	Yes	Currencies Master (P8)	Must reference Active Currency. Cannot be Base Currency.
3	`exchange_rate`	FX Rate (to Base)	معامل التحويل	STORED	Decimal	Yes	-	Precision up to 6 decimals (e.g., 3.750000). > 0.
4	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	-	Inactive blocks non-base transactions for this currency.
5	`updated_by`	Updated By	آخر تحديث بواسطة	STORED	FK	Yes	UserAccount.id	Audit field

6	`updated_at`	Rate Effective Date	تاريخ التحديث	STORED	DateTime	Yes	System	Audit field

9.6.3 Business Rules & Logic

- **Snapshotting for Reports:** When a Tenant pays an invoice in a secondary currency (e.g., USD), the backend immediately reads the current `exchange_rate` from this table and stores it permanently inside the Invoice record. Dashboard reports sum up the calculated base equivalent, completely ignoring future rate fluctuations.

9.7 FX Rate Change Log (سجل تدقيق أسعار الصرف)

9.7.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-09-07`
Form Name	FX Rate Change Log
Form Type	Audit / Read-Only
Purpose	An immutable audit trail documenting every modification made to the exchange rates by Super Admins.

9.7.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Source / Lookup	Validation / Constraints
1	`log_id`	Change ID	معرف التغيير	STORED	Text	Yes	Auto-number	Primary Key
2	`currency_id`	Currency	العملة المعدلة	STORED	FK	Yes	Currencies Master (P8)	Link to the changed currency.
3	`old_rate`	Old FX Rate	السعر القديم	STORED	Decimal	Yes	-	The rate before the update.
4	`new_rate`	New FX Rate	السعر الجديد	STORED	Decimal	Yes	-	The rate after the update.
5	`notes`	Notes	ملاحظات	STORED	Text	No	-	Optional admin notes for the change.
6	`changed_by`	Changed By	المعدل	STORED	FK	Yes	UserAccount.id	Admin who made the change.
7	`changed_at`	Changed At	وقت التعديل	STORED	DateTime	Yes	System	Exact timestamp of the change.

9.7.3 Business Rules & Logic

- **Trigger-Based Creation:** This log is automatically generated by the backend every time an Admin saves a modification in `FRM-CP-09-06`. The Admin cannot manually create or edit entries in this table.

(End of CP-PCS-P9)

Control Panel PCS Part 10

Document ID: CP-PCS-P10 **Title:** Support Management (Centralized Help Desk) **Phase:** 1 (MVP) **Module:** 10. Support Management

10.1 Architectural Strategy & API Bridge Logic

The Centralized Database Concept: To maintain strict database isolation for Tenants while providing a unified Support Inbox for Super Admins, the Support Management module is built entirely within the **Master Admin Database**.

- **Super Admin View:** Operates directly on the Master DB. Can query, filter, and assign any ticket across the entire platform.
- **Tenant View (API Bridge):** The Tenant's isolated database stores **zero** support tickets. When a Tenant navigates to "Support" in their dashboard, their frontend makes a secure RESTful API call to the Master Admin DB (`GET /api/v1/support/tickets`). The backend strictly enforces `WHERE tenant_id = {Current_Tenant}` . This guarantees cross-tenant data privacy without data duplication.

10.2 Support Categories Master (إدارة تصنيفات الدعم)

10.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-10-01`
Form Name	Support Categories Master
Form Type	Configuration (Master Data)
Purpose	Replaces hardcoded ENUMs. Allows Super Admins to dynamically define and manage ticket types (e.g., "Technical Issue", "Billing Inquiry", "Feature Request").

10.2.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Validation / Constraints
1	`category_id`	Category ID	معرف التصنيف	STORED	Text	Yes	Auto-number / UUID (Primary Key)
2	`name_en`	Category Name (EN)	اسم التصنيف (إنجليزي)	STORED	Text	Yes	e.g., "Billing & Invoices"
3	`name_ar`	Category Name (AR)	اسم التصنيف (عربي)	STORED	Text	Yes	e.g., "الفوترة والفواتير"
4	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	Default: True. Inactive categories disappear from Tenant view.

10.3 Canned Responses Master (الردود الجاهزة)

10.3.1 Form Identification Card

Attribute	Description

Form ID	`FRM-CP-10-02`
Form Name	Canned Responses Master
Form Type	Productivity Tool (Admin Only)
Purpose	Allows Support Staff to quickly insert pre-written, standardized replies into tickets, drastically reducing SLA response times.

10.3.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Validation / Constraints
1	`template_id`	Template ID	معرف القالب	STORED	Text	Yes	Auto-number (Primary Key)
2	`title`	Template Title	عنوان القالب	STORED	Text	Yes	e.g., "Requesting Screenshot" (Internal use)
3	`message_body`	Message Content	محتوى الرد	STORED	LongText	Yes	The actual text to be injected into the reply box.
4	`is_active`	Is Active?	مفعل؟	STORED	Boolean	Yes	Default: True.

10.4 Support Ticket Master (Header)

10.4.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-10-03`
Form Name	Support Ticket Master
Form Type	Shared Transactional (Header)
Original Ref.	`SUP-TCK-001`
Purpose	The core entity defining the problem, ownership, and current state.

10.4.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Logic / Constraints
1	`ticket_id`	Ticket ID	المعرف	STORED	UUID	Yes	Primary Key
2	`ticket_no`	Ticket No	رقم التذكرة	STORED	Text	Yes	Auto-sequence (e.g., TKT-10045)

3	`tenant_id`	Tenant	المشترك	STORED	FK	Yes	Link to `Subscriber Profile` (P5).
4	`subject`	Subject	العنوان	STORED	Text	Yes	Short summary of the issue.
5	`category_id`	Category	التصنيف	STORED	FK	Yes	Link to `FRM-CP-10-01`.
6	`priority`	Priority	الأولوية	STORED	Enum	Yes	`Low`, `Medium`, `High`, `Critical`.
7	`status`	Status	الحالة	STORED	Enum	Yes	`New`, `In_Progress`, `Pending_Client`, `Closed`.
8	`assigned_to`	Assigned To	الموظف المسؤول	STORED	FK	No	Links to Admin User. **Invisible to Tenant.**
9	`created_by`	Created By	المنشئ	STORED	Text	Yes	The Tenant User ID who opened it.
10	`created_at`	Created At	تاريخ الإنشاء	STORED	DateTime	Yes	System Time.
11	`closed_at`	Closed At	تاريخ الإغلاق	STORED	DateTime	Cond	System Time when Status becomes `Closed`.

10.5 Ticket Replies (Conversation Thread)

10.5.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-10-04`
Form Name	Ticket Replies
Form Type	Transactional (Child Log)
Original Ref.	`SUP-REP-001`
Purpose	Stores the chronological conversation. Includes architectural modifications for Admin Internal Notes.

10.5.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Logic / Constraints
1	`reply_id`	Reply ID	معرف الرد	STORED	UUID	Yes	Primary Key
2	`ticket_id`	Ticket Ref	التذكرة	STORED	FK	Yes	Link to Parent Ticket (`FRM-CP-10-03`).
3	`sender_type`	Sender Type	نوع المرسل	STORED	Enum	Yes	`Tenant_User`, `Admin_Support`, `System_Bot`.
4	`sender_id`	Sender ID	المرسل	STORED	Text	Yes	The ID of the person/bot who wrote it.

5	`message_body`	Message	نص الرسالة	STORED	LongText	Yes	Rich text.
6	`attachment_url`	Attachment	المرفق	STORED	File	No	Single file upload URL (Image/PDF).
7	`is_internal`	Internal Note?	ملاحظة داخلية؟	STORED	Boolean	Yes	Default: False. **If True, entirely hidden from Tenant API.**
8	`created_at`	Timestamp	الوقت	STORED	DateTime	Yes	System Time. Determines sorting order.

10.6 Operational Workflows & Automated State Machine

10.6.1 Automated Status Transitions (Ping-Pong Logic)

To ensure accurate SLA tracking and reduce manual administrative work, the Ticket `status` changes automatically based on reply triggers:

- **Trigger 1:** A new ticket is created ➔ Status = `New`.
- **Trigger 2:** Admin User assigns themselves or is assigned ➔ Status = `In_Progress`.
- **Trigger 3:** Admin User adds a reply (where `is_internal` = False) ➔ Status = `Pending_Client`.
- **Trigger 4:** Tenant User adds a reply to an open ticket ➔ Status = `In_Progress` (This instantly flags the ticket on the Admin Dashboard as requiring attention).

10.6.2 Role-Based Access Control (RBAC) & View Restrictions

- Tenant Limitations:
  - Cannot view or edit `assigned_to`.
  - Cannot view any reply where `is_internal == True`.
  - Cannot change the `priority` after the ticket is created.
  - Cannot change status to `Closed`. Can only request closure or reply.
  - **Strict Security:** Cannot add a reply if the Ticket `status == Closed`. They must open a new ticket.
- Admin Superpowers:
  - Can view all tickets.
  - Can override and change `priority` (e.g., downgrading a false "Critical" ticket to "Low").
  - Can change `assigned_to` to route tickets to specialized departments (e.g., from Technical to Billing).
  - Can force close a ticket at any time.

(End of CP-PCS-P10)

Control Panel PCS Part 11

Document ID: CP-PCS-P11 **Title:** Security, Audit Logs & Sessions **Phase:** 1 (MVP) **Module:** 11. Security & Audit

11.1 Admin Security Settings (إعدادات أمان الإدارة)

11.1.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-11-01`
Form Name	Admin Security Settings
Form Type	Configuration (Single Record)
Purpose	Strictly isolates and manages security policies for Super Admins and IRoad internal staff.

11.1.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Logic / Validation
1	`setting_id`	Setting ID	المعرف	STORED	Text	Yes	Fixed ID (e.g., "ADMIN-SEC-CONF").
2	`session_timeout_minutes`	Admin Session TTL (Mins)	مدة الجلسة (دقائق)	STORED	Integer	Yes	Default: 240 (4 Hours). Forces re-login.
3	`max_failed_logins`	Max Failed Attempts	أقصى محاولات فاشلة	STORED	Integer	Yes	Default: 3.
4	`lockout_duration_minutes`	Lockout Duration (Mins)	مدة الحظر (دقائق)	STORED	Integer	Yes	Default: 30.
5	`updated_by`	Updated By	آخر تحديث بواسطة	STORED	FK	Yes	Admin User ID.
6	`updated_at`	Updated At	وقت التحديث	STORED	DateTime	Yes	System Audit.

11.2 Tenant & External Security Settings (إعدادات أمان المشتركين)

11.2.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-11-02`
Form Name	Tenant Security Settings

Form Type	Configuration (Single Record)
Purpose	Strictly isolates and manages security policies for external users (Tenants' staff and their Drivers).

11.2.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Logic / Validation
1	`setting_id`	Setting ID	المعرف	STORED	Text	Yes	Fixed ID (e.g., "TENANT-SEC-CONF").
2	`tenant_web_timeout_hours`	Web Session TTL (Hours)	جلسة المشترك (ساعات)	STORED	Integer	Yes	Default: 12. For Tenant dashboard users.
3	`driver_app_timeout_days`	Driver App TTL (Days)	جلسة السائق (أيام)	STORED	Integer	Yes	Default: 30. Mobile app JWT expiry.
4	`max_failed_logins`	Max Failed Attempts	أقصى محاولات فاشلة	STORED	Integer	Yes	Default: 5.
5	`lockout_duration_minutes`	Lockout Duration (Mins)	مدة الحظر (دقائق)	STORED	Integer	Yes	Default: 15.
6	`updated_by`	Updated By	آخر تحديث بواسطة	STORED	FK	Yes	Admin User ID.
7	`updated_at`	Updated At	وقت التحديث	STORED	DateTime	Yes	System Audit.

11.3 Active Sessions Monitor (مراقب الجلسات الحية)

11.3.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-11-03`
Form Name	Active Sessions Monitor
Form Type	Operational Dashboard (Read/Action)
Purpose	Live monitoring of all active JWTs across the platform. Empowers Admins to instantly kill compromised or unwanted sessions.

11.3.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Logic / Validation
1	`session_id`	Session ID	معرف الجلسة	CACHE	Text	Yes	Redis key or JWT JTI.
2	`user_domain`	User Domain	نطاق المستخدم	CACHE	Enum	Yes	`Admin`, `Tenant_User`, `Driver`.

3	`reference_id`	User Reference	المعرف	CACHE	FK	Yes	Link to the specific user/driver.
4	`ip_address`	IP Address	عنوان IP	CACHE	Text	Yes	Captured upon login.
5	`user_agent`	Device / Browser	الجهاز المتصل	CACHE	Text	Yes	Browser or App version.
6	`started_at`	Started At	وقت البدء	CACHE	DateTime	Yes	Session initiation time.

11.3.3 Business Rules & Kill Switch Logic

- **Targeted Revocation:** Admin can click "Revoke" on a specific `session_id` to instantly block that specific device.
- **Domain Revocation (The Kill Switch):** Admin can input a `Tenant_ID` and execute a "Mass Revoke", instantly destroying all active sessions where `reference_id` belongs to that Tenant.

11.4 Authentication & Access Log (سجل محاولات الدخول)

11.4.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-11-04`
Form Name	Access Log
Form Type	Immutable Audit Log (Read-Only)
Purpose	Tracks every login/logout attempt across all domains to detect brute-force attacks and unauthorized access.

11.4.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Logic / Validation
1	`log_id`	Log ID	المعرف	STORED	Auto	Yes	Primary Key.
2	`attempt_type`	Attempt Type	نوع المحاولة	STORED	Enum	Yes	`Login`, `Logout`, `Token_Refresh`.
3	`status`	Status	الحالة	STORED	Enum	Yes	`Success`, `Failed`, `Blocked`.
4	`user_domain`	Domain	النطاق	STORED	Enum	Yes	`Admin`, `Tenant_User`, `Driver`.
5	`email_used`	Email Used	البريد المستخدم	STORED	Text	Yes	Captured even if the user doesn't exist.
6	`ip_address`	IP Address	عنوان IP	STORED	Text	Yes	Source of the attempt.
7	`timestamp`	Timestamp	الوقت	STORED	DateTime	Yes	Exact time of attempt.

11.5 Master System Audit Log (الصندوق الأسود للتدقيق)

11.5.1 Form Identification Card

Attribute	Description
Form ID	`FRM-CP-11-05`
Form Name	Master System Audit Log
Form Type	Immutable Audit Log (Read-Only)
Purpose	The ultimate accountability tracker. Records strictly "What the Admins did" inside the Control Panel to prevent untracked configuration or financial changes.

11.5.2 Main Fields (Data Dictionary)

#	Field Key	Display Label (EN)	Display Label (AR)	Class	Data Type	Req.	Logic / Validation
1	`audit_id`	Audit ID	رقم السجل	STORED	Auto	Yes	Primary Key.
2	`admin_id`	Performed By	منفذ الإجراء	STORED	FK	Yes	Link to Admin User.
3	`action_type`	Action	الإجراء	STORED	Enum	Yes	`Create`, `Update`, `Delete`, `Status_Change`.
4	`module_name`	Module / Form	الوحدة	STORED	Text	Yes	e.g., "Tax Settings", "Subscription Plans".
5	`record_id`	Affected Record	السجل المتأثر	STORED	Text	Yes	The ID of the modified entity.
6	`old_payload`	Old Values (JSON)	القيم القديمة	STORED	JSON	No	Null if `Create`. Contains state before update.
7	`new_payload`	New Values (JSON)	القيم الجديدة	STORED	JSON	No	Null if `Delete`. Contains state after update.
8	`ip_address`	IP Address	عنوان IP	STORED	Text	Yes	Admin's IP at the time of modification.
9	`timestamp`	Timestamp	وقت الإجراء	STORED	DateTime	Yes	System Time.

11.5.3 Business Rules & Logic

- **Absolute Immutability:** The database user/role connecting the backend to this table must ONLY have `INSERT` and `SELECT` privileges. `UPDATE` and `DELETE` commands must be strictly disabled at the database engine level to guarantee forensic integrity.

(End of CP-PCS-P11)

Control Panel PCS Part 12

Document ID: CP-PCS-P12 **Title:** System Intersection Points & Shared Services (Admin vs. Tenant) **Phase:** 1 (MVP) **Module:** 12. Architectural Intersections

12.1 Overview & Architectural Philosophy

In the IRoad Multi-Tenant architecture, the **Super Admin Environment** (Central Master DB) and the **Tenant Environment** (Isolated DB) are strictly separated to guarantee data privacy and operational autonomy. However, for the SaaS platform to function cohesively, strategic "Bridges" or "Intersection Points" must exist.

This document defines how these shared services operate logically, defining the boundaries of authority, the flow of data, and the strict technical mechanisms (e.g., Event Buses, Replication, API Bridges) used to prevent tight database coupling and ensure referential integrity.

12.2 Global Master Data (Data Standardization & FK Integrity)

Intersection Type: Automated Read-Only Replication (Seed & Sync)

- **The Challenge:** Fetching global data (Countries, Currencies) purely via API for UI dropdowns breaks database Referential Integrity. A Tenant's local invoice cannot have a Foreign Key pointing to a currency code that doesn't exist in their isolated database.
- **How it works:** The Super Admin possesses absolute authority to create and manage global datasets. When a new Tenant is provisioned, the system **seeds** a read-only copy of these tables into the Tenant's isolated DB. If the Super Admin updates a currency or adds a country later, the Master system publishes a synchronization event (via Internal Event Bus) that updates the local tables across all active Tenant databases instantly.
- **Tenant Experience:** Can only view and select data. Cannot add or modify countries/currencies.
- **Architectural Value:** Ensures strict database Referential Integrity (Foreign Key constraints) within the Tenant's environment while preventing data pollution and maintaining standardized ISO codes globally.

12.3 Organization Profile vs. Subscriber Profile (CRM)

Intersection Type: Event-Driven Two-Way Synchronization

- **The Challenge:** Direct database cross-talk between the Tenant DB and Master DB to update a profile causes deadlocks and violates the isolation principle.
- **How it works:** Both entities refer to the same logical company but are presented asymmetrically. Synchronization is handled via an **Internal Event Bus (Message Broker/Webhooks)**.
  - *Tenant Updates:* When the Tenant changes their phone number in their "Organization Profile", the Tenant DB fires a `Tenant_Profile_Updated` event. The Master DB consumes this event and updates the "Subscriber CRM Profile" asynchronously.
  - *Admin Exclusive Fields:* The Master CRM contains fields (LTV, Wallet Balance, Account Status) that are strictly isolated and never transmitted back to the Tenant DB.
- **Architectural Value:** Achieves real-time data mirroring for the Sales Team without establishing dangerous direct database links, ensuring maximum system resilience.

12.4 Centralized Help Desk (Support Tickets)

Intersection Type: Shared Collaborative Workspace (API Bridge)

- **How it works:** To prevent the IRoad Support Team from logging into hundreds of isolated databases, the Support Ticketing system physically resides entirely within the Master Admin DB.
- **Tenant Experience:** Operates through a secure API bridge. The Tenant opens the "Support" tab, but their frontend queries the Master Admin API (`GET /tickets?tenant_id=XYZ`). They can create new tickets and reply to open threads.

- **Admin Experience:** The Admin possesses a global inbox viewing all tickets across the platform. They hold "Superpowers": the ability to override ticket priorities, reassign tickets, and write **Internal Notes** (Hidden replies visible only to Admin staff, completely masked from the Tenant API).
- **Architectural Value:** Delivers a centralized, highly efficient operational hub for the SaaS operator while maintaining the illusion of an embedded support module for the Transporter.

12.5 Subscription Billing & Resource Provisioning

Intersection Type: Governed Self-Service & Asynchronous Workflows

- **How it works:** This intersection bridges the financial authority of the Admin with the operational needs of the Tenant, utilizing two distinct operational workflows.
- **A. The Automated Bridge (Online Gateways):** The Tenant selects a plan and pays via Stripe/Moyasar. Upon a successful Webhook from the gateway, the Master System automatically generates the Immutable Standard Invoice and immediately executes a provisioning command to update the operational limits (e.g., Max Trucks) inside the Tenant's isolated database.
- **B. The Manual Approval Bridge (Offline Bank Transfer):** The Tenant selects "Bank Transfer" and uploads a receipt. The order is placed in a `Pending_Payment` state in the Master DB. The Tenant's limits remain unchanged. Once the Super Admin reviews the receipt and clicks "Approve", the system triggers the internal provisioning command to unlock the Tenant's resources.
- **Architectural Value:** Creates a frictionless revenue engine with built-in safeguards for manual auditing, ensuring resources are never provisioned without verified financial settlement.

12.6 Cross-Environment Tax & Rules Engine

Intersection Type: Automated Read-Only Replication

- **The Challenge:** A Tenant needs to issue Sales Invoices to their own clients daily. If their system must query the Admin Master DB to calculate taxes for every invoice, the latency will be high, and an Admin server outage would paralyze all Tenants' commercial operations.
- **How it works:** Similar to Global Master Data, the Super Admin defines the "Tax Codes Master" centrally (e.g., VAT 15% for SA, Zero-rate for International). This logic is **replicated locally** to every Tenant's database.
- **Architectural Value:** Allows the Tenant's local invoicing engine to perform complex tax routing and financial calculations instantly, with zero latency and 100% uptime, completely independent of the Master DB's live availability.

12.7 Communication Engine & System Banners

Intersection Type: Top-Down Broadcast & Centralized Dispatch

- **How it works:** The communication infrastructure (SMTP and SMS gateways) is centralized in the Master DB to optimize API costs and maintain brand consistency.
- **Tenant Event Triggers:** When a Tenant action requires an alert (e.g., OTP request), their isolated DB sends an API dispatch request to the Admin's Central Engine. The Admin engine utilizes predefined Notification Templates to format and send the message.
- **Top-Down Broadcasting (Banners):** The Admin can create a "System Banner" (e.g., "Scheduled Maintenance"). This banner is fetched by the Tenant's UI upon login and displayed globally.
- **Architectural Value:** Prevents Tenants from tampering with system-critical communication templates and gives the platform operator a unified voice.

12.8 Security, Sessions & The "Kill Switch"

Intersection Type: Invisible Enforcement & Absolute Override

- **How it works:** Security parameters are established centrally but executed globally across all domains via the caching layer (Redis).
- **Passive Enforcement:** The Admin dictates global security policies (e.g., Tenant web sessions expire after 12 hours). The authentication service enforces these rules.
- **Active Override (The Kill Switch):** If a Tenant defaults on payment or commits a severe violation, the Admin changes their CRM status to `Suspended`.
- **The Execution:** This status change fires a critical platform event. The identity provider immediately targets the Redis cache and destroys/revokes all active JWTs linked to that `Tenant_ID`.

- **Result:** Every employee, dispatcher, and driver belonging to that specific Tenant is instantly and forcefully logged out of all web and mobile applications.
- **Architectural Value:** Guarantees IRoad's absolute sovereignty and commercial leverage over platform access, mitigating financial and security risks instantaneously.

(End of CP-PCS-P12)